

13. (1.5 points)

An actuary is reviewing residual plots from an over-dispersed Poisson bootstrapping model. The actuary chooses to review a plot of the residuals vs. development periods.

a. (0.5 point)

Identify two other residual plots the actuary might choose to review.

b. (0.5 point)

Briefly describe two features of residual plots that would suggest a need for the actuary to adjust the model.

c. (0.5 point)

In reviewing the plot of residuals vs. development periods, the actuary notices that the residuals appear to have larger absolute values at lower maturities. The actuary argues this is to be expected, because the incremental values are much larger in the earlier development periods and hence these incremental values should have a higher variance.

Assess the validity of the actuary's reasoning.

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EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 13	
TOTAL POINT VALUE: 1.5	LEARNING OBJECTIVE: A9: Test assumptions underlying reserve models.
SAMPLE ANSWERS (BY PART, AS APPLICABLE)	
Part a: 0.5 point	
<i>Sample Answers (any two of which would earn full credit)</i> • Residuals vs. Accident Year • Residuals vs. Calendar Year • Residuals vs. Size of Loss (Prior Cumulative, Expected Incremental, etc.) • Normality Plot • Box and Whisker Plot	
Part b: 0.5 point	
<i>Sample Answer 1</i> 1) If the actuary notices that variance is not constant across all residuals. 2) If the actuary notices that residuals are trending, so for example, early AYs have positive residuals and later AYs have negative residuals. <i>Sample Answer 2</i> 1) Downward trend in residuals 2) Residuals become more dispersed (or less dispersed) at different parts of the plot <i>Sample Answer 3</i> 1) If residuals become more spread out or have varying dispersion among periods, suggesting heteroscedasticity. <i>Sample Answer 4</i> 1) If residuals become more spread out or have varying dispersion among periods, suggesting heteroscedasticity. <i>Sample Answer 5</i> 1) Outliers in residuals We also would have accepted: The residuals in the normality plot are not tightly grouped around a straight 45 degree line through the origin.	
Part c: 0.5 point	
<i>Sample Answer 1</i> His reasoning is not sound. Each residual is divided by the square root its expected variance based on the ODP model. Therefore if there is still variation in spread of residuals we have unexpected changes in variance and need to make an adjustment to our model. <i>Sample Answer 2</i> This could be accurate if they are raw residuals. However, if they are weighted-residuals, we would not expect to see this.	

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

EXAMINER'S REPORT (BY PART, AS APPLICABLE)
Part a
Candidates performed very well on this part. They were expected to identify two residuals plots, other than the one mentioned in the stem, that an actuary could choose to review. Common errors were describing residuals vs. age (as this was the example given in the question), and describing plots that do not have residuals (e.g., p-p plot).
Part b
Candidates performed well on this part. Candidates were expected to describe specific visual features of a plot that would indicate an adjustment was needed. Common errors included describing the same feature twice on different plots, describing the adjustment needed without the plot features (e.g. simply saying there was heteroscedasticity in the plot), and generally vague descriptions. Another common error was describing zero sum residuals, as this would not be identified through a plot. Although not required, many candidates created a sample plot which made it easier to interpret their intent.
Part c
Candidates' performance on this part was more mixed. Candidates were expected to directly address the argument made, reach a conclusion and provide a rationale.
Many candidates do not address the argument at all and simply stated that the bootstrap model requires residuals to be iid – this response was awarded no credit as it failed to address the actuary's reasoning as instructed in the question.
Other common mistakes included commenting on the variability at different ages due to claim development and agreeing that the actuary was correct.